

Code

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 from sklearn.model_selection import train_test_split
4 from sklearn.linear_model import LinearRegression
5 from sklearn.metrics import mean_squared_error, r2_score
6
7 # Load Dataset
8 df = pd.read_csv('/content/Advertising.csv')
9
10 # Remove unwanted index column if present
11 if 'Unnamed: 0' in df.columns:
12     df = df.drop('Unnamed: 0', axis=1)
13
14 # Features and Target
15 X = df[['TV', 'Radio', 'Newspaper']]
16 y = df['Sales']
17
18 # Split Dataset
19 X_train, X_test, y_train, y_test = train_test_split(
20     X, y, test_size=0.2, random_state=42
21 )
22
23 # Train Model
24 model = LinearRegression()
25 model.fit(X_train, y_train)
26
27 # Prediction
28 y_pred = model.predict(X_test)
29
30 # Evaluation
31 print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
32 print("R2 Score:", r2_score(y_test, y_pred))
33
34 # Actual vs Predicted Graph
35 plt.figure(figsize=(8,5))
36 plt.plot(y_test.values, label='Actual Sales')
37 plt.plot(y_pred, label='Predicted Sales')
38 plt.xlabel("Data Points")
39 plt.ylabel("Sales")
40 plt.title("Sales Prediction")
41 plt.legend()
42 plt.show()
```

Output

```
Mean Squared Error: 3.1740973539761033
R2 Score: 0.899438024100912
```

Plot

